Chapter 12 The Laws of Thermodynamics

Definition of Thermodynamic

The study of heat and its relationship to work, energy and matter.

Internal Energy

Represent the total energy contained within a system, encompassing kinetic and potential energies of its particles. (U)

Heat Q and Work W

Heat is the transfer of energy due to a temperature difference.
Work is the transfer of energy due to force acting through a distance.

$$W = Fx = pSx = p\Delta V$$

The First Law Equation

$$\Delta U = Q - W$$

ΔU is the change in interna energy, Q is the heat added to the system, W is the work done by the system.

When temperature increases, ΔU is positive.

The Law provides a connection between microscopic and macroscopic worlds. And changes in the internal energy result in changes in the measurable macroscopic variables of the system

- pressure
- temperature
- volume

Work in Thermodynamic Processes

Assumptions:

- Deal with ideal gas
- In thermodynamic equilibrium(same temperature and pressure of every part of the gas)

And

$$W = P\Delta V$$

can comprehend the energy changes.

Second Law of Thermodynamics

Clausius:
heat cannot spontaneously transfer from a lower temperature body to a higher temperature body.
Kelvin:
it is impossible to extract heat from a single source and convert it completely into useful work without causing other side effects.
The principle of entropy increases:
the increase in entropy during an irreversible thermodynamic process is always greater than zero.

It shows the direction of natural processes and the behavior of energy with in systems.
If heat are made to flow in the opposite direction, it need external work. This means that the Second Law limits the efficiency of energy transformations.

The Second Law also shows the direction of time.

Entropy and Disorder

Entropy is a measure of disorder or randomness in a system. The Second law shows that natural processes tend to move towards greater disorder.
Spontaneous process in an isolated system always increase entropy.

Methods of Heat Transfer

Need to know the rate and mechanisms of heat transfer.

Methods include

- conduction
- convection(热对流)
- radiation

Conduction

It is an exchange of energy between microscopic particles by collisions and less energetic particles gain energy during collisions with more energetic particles.
Rate depends on the substance.

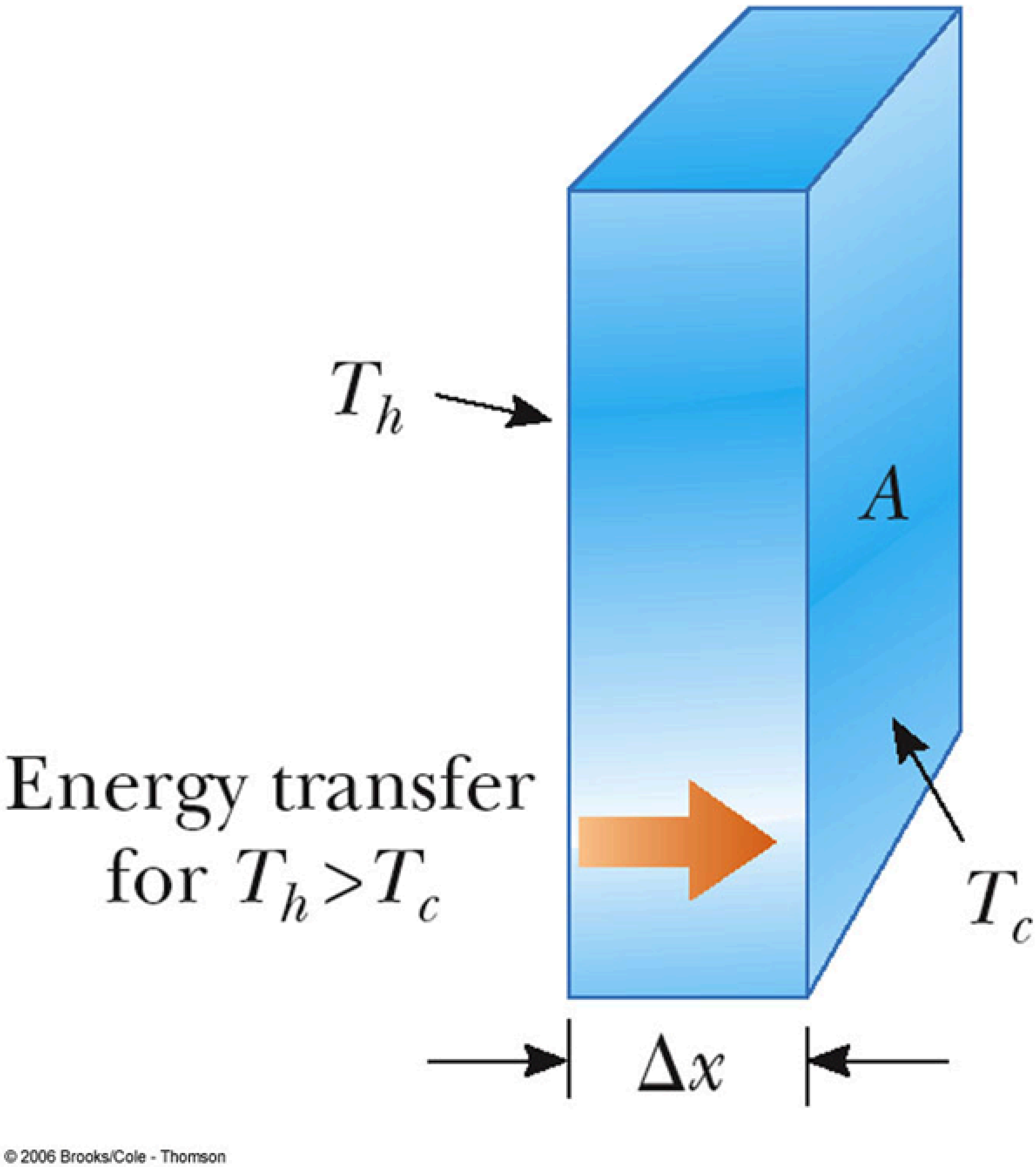
In general, metals are good conductors, which contain large number of free electrons and transport energy.

Conduction can occur only if there is a difference in temperature between two parts of conducting medium.

Conduction Equation

$$P_{\text{cond}} = \frac{Q}{t} = kA\frac{T_H - T_C}{L}$$

where P_{cond} is the rate of conduction, k is thermal conductivity, A is the area of conduction, T_H is the high temperature and T_c is low temperature.



For slab, the heat resistance

$$R = \frac{L}{kA}$$

Thus the P_{cond} is similar to I, ΔT is similar to U. The heat resistance is similar to electric resistance.
For steady state, the energy transfer rate is the same in each layer.

Convection

Energy transferred by the movement of a substance.
When the movement results from differences in temperature, it is called natural convection;
When the movement is forced, it is called forced convection.

Example

Air directly above the flame is warmed and rises, transferring the energies.

Applications:

- boiling water
- radiators
- cooling automobile engines

Radiation

It does not require physical contact. Energy transferred by electromagnetic waves due to thermal vibrations of molecules.

Radiation Equation

$$P_{\text{rad}} = \sigma \epsilon A T^4$$

where P_{rad} is the rate of energy transfer, A is the surface area of the object, σ is a constant, ϵ is emissivity(vary from 0 to 1) and T is Kelvin temperature.

In general, an object also absorb the energy by radiation. Thus

$$P_{\text{net}} = P_{\text{abs}} - P_{\text{rad}} = \sigma \epsilon A (T_{\text{env}}^4 - T^4)$$

When an object is in equilibrium, it radiates and absorbs at the same rate.

Ideal Absorbers

A ideal absorber is an object that absorbs all of the energy incident on it. $\epsilon = 1$. Also called the black body or ideal radiator. A ideal reflector absorbs none of the energy incident on it. $\epsilon = 0$.

Ideal Gas

- No fixed volume and pressure and can expand to fill the container.

- Most gases at room temperature and pressure behave approximately as an ideal gas.
- Atoms or molecules move randomly
- No long-range force on one another
- Each particle is point-like

Moles

Used to express the amount of gas in a given volume.

$$n = \frac{m}{M}$$

where m is mass and M is molar mass.

Avogadro's Number

The number of particles in a mole is called Avogadro's Number.
The mass of an individual atom

$$m_{\text{atom}} = \frac{M}{N_A}$$

Ideal Gas Law

$$PV = nRT$$

where R is the universal gas constant, n is the number of moles, T is Kelvin temperature, P is pressure and V is volume.

Alternatively,

$$PV = Nk_BT$$

where k_B is Boltzmann's constant and $k_B = \frac{R}{N_A}$. N is the total number of molecules.

Kinetic Theory of Gases

Assumptions:

- Large number of molecules, average separations is larger than their dimensions.
- Obey Newton's Laws of motion, move randomly as a whole
- Interact only by short-range forces during elastic collisions
- Molecules elastically collide with the wall.
- Gas is pure substance, all molecules are identical.

Pressure of an Ideal Gas

The pressure is proportional to the number of molecules per unit volume and to the average translational kinetic energy of a molecule.

Average Translational Kinetic Energy

$$K_{\text{avg}} = \frac{3}{2}kT, k = \frac{R}{N_A}$$

That is, at a given temperature, all ideal gas molecules has the same average translational kinetic energy.

Internal Energy

For monatomic gas, the KE is the only type of energy the molecules can have

$$U = \frac{3}{2}nRT$$

Thus

$$\Delta U = \frac{3}{2}nR\Delta T$$

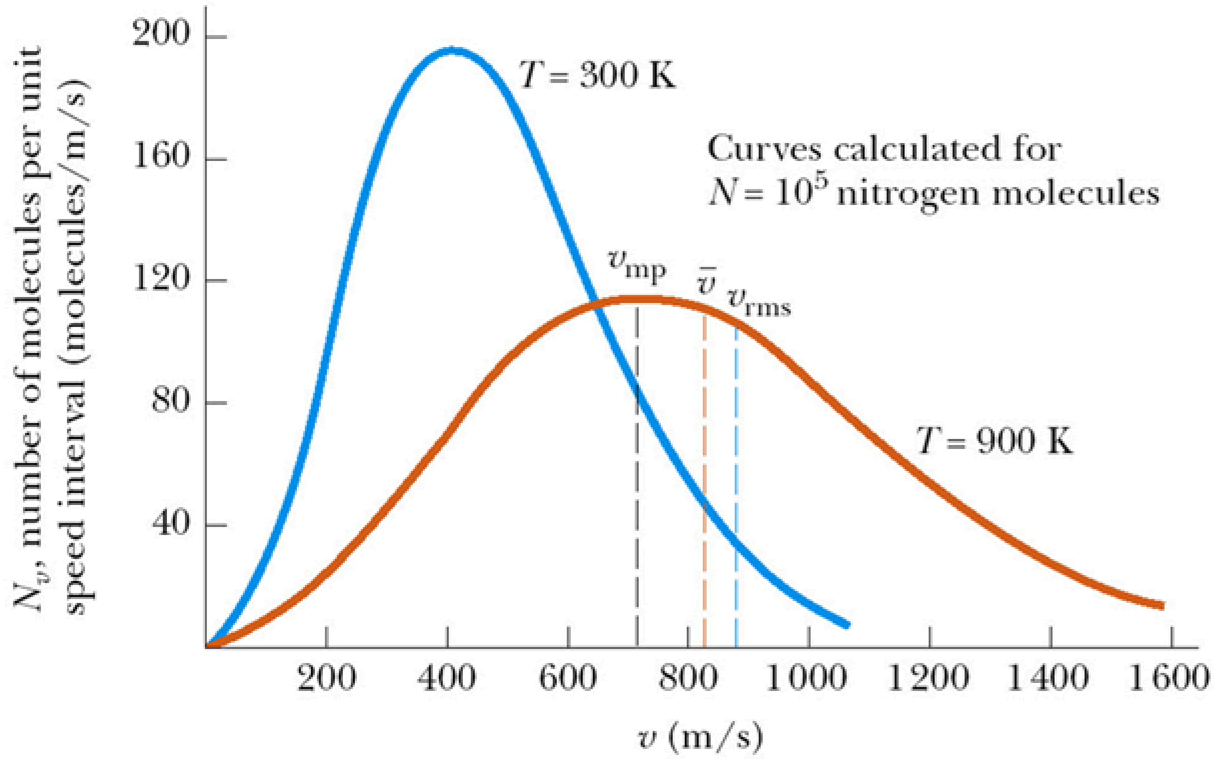
Speed of Molecules

$$v_{\text{avg}} = \sqrt{\frac{8RT}{\pi M}}$$
$$v_{\text{P}} = \sqrt{\frac{2RT}{M}}$$
$$v_{\text{rms}} = \sqrt{\frac{3RT}{M}}$$

Average speed, most probable speed, rms speed.(root-mean-square)

Maxwell Distribution

A system of gas at a given temperature will exhibit a variety of speeds. When temperature increases, the curve move to the region of high speed, the region of distribution widen.


N_v, number of molecules per unit speed interval (molecules/m/s)
200
160
120
80
40
$T = 300$ K
Curves calculated for
$N = 10^5$ nitrogen molecules
v_{mp}
$\bar{v}$
v_{rms}
$T = 900$ K
200
400
600
800
1000
1200
1400
1600
v (m/s)